It is clear that whenever a constant is added or subtracted with the independent variable x, the fundamental formulae remain the same.

But

$$\frac{d}{dx}\left[\frac{1}{l}(e^{lx+m})\right] = e^{lx+m} \Rightarrow \int e^{lx+m}dx = \frac{1}{l}e^{(lx+m)} + c$$

$$\frac{d}{dx}\left[\frac{1}{a}\sin(ax+b)]\right] = \cos(ax+b) \Rightarrow \int \cos(ax+b)dx = \frac{1}{a}\sin(ax+b) + c$$

Here, if any constant is multiplied with the independent variable x, then the same fundamental formula can be used after dividing it by the coefficient of x

That is, if $\int f(x)dx = g(x) + c$, then $\int f(ax+b)dx = \frac{1}{a}g(ax+b) + c$

The above formula can also be derived by using substitution method, which will be studied later.

Example 11.4

Evaluate the following with respect to x:

(i) $\int (4x+5)^6\, dx$ (ii) $\int \sqrt{(15-2x)}\, dx$ (iii) $\int \frac{1}{(3x+7)^4} dx$

Solution

(i) $\int (4x+5)^6\, dx = \frac{1}{4}\frac{(4x+5)^{6+1}}{6+1} = \frac{(4x+5)^7}{28} + c$

(ii) $\int \sqrt{(15-2x)}\, dx = \int (15-2x)^{\frac{1}{2}}\, dx = \left(\frac{1}{-2}\right)\frac{(15-2x)^{\frac{1}{2}+1}}{(1/2)+1} = -\frac{(15-2x)^{\frac{3}{2}}}{3} + c$

(iii) $\int \frac{1}{(3x+7)^4} dx = \int (3x+7)^{-4}\, dx = \frac{1}{3}\frac{(3x+7)^{-4+1}}{-4+1} = -\frac{1}{9(3x+7)^3} + c$

Example 11.5

Integrate the following with respect to x:

(i) $\sin(2x + 4)$ (ii) $\sec^2(3+4x)$ (iii) $\text{cosec}(ax+b)\cot(ax+b)$

Solution

(i) $\int \sin(2x + 4)\, dx = \left(\frac{1}{2}\right)(-\cos(2x+4)) + c = -\frac{1}{2}\cos(2x+4) + c$

(ii) $\int \sec^2(3+4x)\, dx = \frac{1}{4}\tan(3+4x) + c$

(iii) $\int \text{cosec}(ax+b)\cot(ax+b)\, dx = \left(\frac{1}{a}\right)(-\text{cosec}(ax+b)) + c = -\frac{1}{a}\text{cosec}(ax+b) + c$

Example 11.6

Integrate the following with respect to x:

(i) e^{3x} (ii) e^{5-4x} (iii) $\frac{1}{(3x-2)}$ (iv) $\frac{1}{(5-4x)}$

Solution

(i) $\int e^{3x}\, dx = \frac{1}{3}e^{3x} + c$ (ii) $\int e^{5-4x}\, dx = -\frac{e^{5-4x}}{4} + c$

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